Bingxuan Li

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Research Interests

My research lies at the intersection of deep learning and computer graphics, with a focus on efficient visual representations, 3D understanding, and controllable generation of visual content and motion.

Education

New York University — New York City, USA
Doctor of Philosophy in Computer Science — *Sep 2024 – Present*
Advisor: Prof. Qi Sun

Peking University — Beijing, China
Bachelor of Science in Computer Science with Honours, Turing Class — *Sep 2020 – Jun 2024*

Work Experience

Research Intern, TikTok — San Jose, USA
Improved motion generation quality through post-training techniques. — *May 2026 – Aug 2026*
Mentors: Dr. Yili Zhao, Dr. Siqi Wang, Xinhao Song

Research

- **Frequency-Adaptive Gaussian Grids for Efficient Image Representation**
  WACV 2027 | To appear
  Anatolii Evdokimov, Stavros Diolatzis, **Bingxuan Li**, Akshay Jindal, Patsorn Sangkloy, Qi Sun
- **Physically Grounded Monocular Depth via Nanophotonic Wavefront Encoding**
  ECCV 2026 | Paper
  Bingxuan Li*, Jiahao Wu*, Yuan Xu*, Zezheng Zhu, Yunxiang Zhang, Kenneth Chen, Yanqi Liang, Nanfang Yu†, Qi Sun†
- **Nano-Optics for Depth Sensing**
  ACM SIGGRAPH 2025 Emerging Technologies | Paper
  Yuan Xu*, **Bingxuan Li***, Jiahao Wu*, Yunxiang Zhang, Zezheng Zhu, Nanfang Yu†, Qi Sun†
- **Image-GS: Content-Adaptive Image Representation via 2D Gaussians**
  ACM SIGGRAPH 2025 | Paper | Project | Video
  Yunxiang Zhang*, **Bingxuan Li***, Alexandr Kuznetsov, Akshay Jindal, Kenneth Chen, Anton Sochenov, Anton Kaplanyan, Qi Sun†
- **Proxy Tracing: Unbiased Reciprocal Estimation for Optimized Sampling in BDPT**
  ACM Transactions on Graphics (ACM SIGGRAPH 2024) | Paper | Project | Video
  Fujia Su*, **Bingxuan Li***, Qingyang Yin, Yanchen Zhang, Sheng Li†

  * Equal contributions   † Corresponding authors

Awards

New York University	SoE Fellowship (2024)
Peking University	John Hopcroft Scholarship (2023)
Peking University	John Hopcroft Scholarship (2021)
Peking University	Freshman Scholarship (2020)
Chinese Chemistry Olympiad	Silver Medal (2018)

Skills

- **Programming**: Python, C/C++
- **Tools**: PyTorch, CUDA
- **Language**: Mandarin, English